GDD

Conceptual Models

Cognitive Model

Attribute	Value	Explanation
Self-Esteem	0-1 (Float Range)	Represents NPC's internal sense of worth or confidence. High values indicate confident decision-making; low values create vulnerability to social influence or self-doubt
Locus of Control	0-1 (Float Range)	Measures perceived control over outcomes. 0 = external (blames fate/others), 1 = internal (believes actions determine outcomes). Influences responsibility attribution in dialogue and action.
Cognitive Flexibility	0-1 (Float Range)	Ability to adapt beliefs or strategies when presented with new information. Low values = rigid, dogmatic thought; high values = adaptive, open to persuasion or compromise.

Social Personality Model

Attribute	Value	Explanation
Assertion	0-1 (Float Range)	0 = highly passive (Easily persuaded), 1 = highly assertive (sticks to their opinion, more likely to

		challenge player). Determines how NPC balances ideology vs practical solutions.
Conformity / Independence	0-1 (Float Range)	0 = high conformity (follows group norms), 1 = high independence (acts autonomously). Influences alignment with factions, responsiveness to social cues, and willingness to challenge authority.
Empathy / Self Focus	0-1 (Float Range)	0 = highly self-focused, 1 = highly empathetic. Governs responsiveness to others' emotions and how relationally dialogue is interpreted.
Ideology	Enum	Broad guiding principle or worldview (e.g., Utilitarianism, Socialism, Hedonism). Impacts decision-making, long-term goals, and dialogue priorities.
Wildcard	Enum	Psychological modifier representing deep-seated complexes, obsessions, or neuroses (e.g., Guilt Complex, Napoleon Complex, Inferiority Complex). Can trigger context-specific behavioral deviations, dialogue modifiers, or unique narrative choices.
Faction	Enum	Political, social, or factional affiliation (e.g., I***nsurgent Militia, Commonman***

		Artists, Runner Network). Determines social context, alliances, and faction-driven goals.
Social Position	0-1 (Float Range)	rank or influence within their faction, determining authority, decision-making power, and how they interact with others, including the player. ***(Boss, Vice, Higher, Member.)***

World Perception Model

Attribute	Value	Explanation
World History	string	Full objective summary of the world's events, factions, wars, and politics. Serves as the reference for NPC understanding of large-scale events.
Personal History	string	Objective record of NPC-specific experiences (milestones, personal achievements, trauma). Acts as grounding for subjective interpretation of events.
Player History	string	Record of all observable player actions (quests completed, alliances, choices). Subjectively interpreted through the NPC's cognitive and social model to create emergent dialogue responses.
Relation	0-1 (Float)	How much respect the player and the NPC have built with eachother. 0 Represents contempt, 1

		indicates Close friendship. 0.5 Is Neutrality.

Examples - Main NPCS

Morisson Moses

Cognitive Model

Attribute	Value
Self-Esteem	0.8
Locus of Control	0.475
Cognitive Flexibility	0.3

Social Personality Model

Attribute	Value
Assertion	1
Conformity / Independence	0.7
Empathy / Self Focus	0.45
Ideology	Utilitarianism
Wildcard	Martyr
Faction	Insurgency
Social Position	Boss
Relation	X

Historical Model

Attribute	Value
World History	Check Below
Personal History	N/A
Player History	N/A

Amourie Othella

Cognitive Model

Attribute	Value
Self-Esteem	0.9
Locus of Control	0.2
Cognitive Flexibility	0.8

Social Personality Model

Attribute	Value
Assertion	0.5
Conformity / Independence	0.6
Empathy / Self Focus	0.25
Ideology	Communitarianism
Wildcard	Napoleon
Faction	The Commonman
Social Position	Boss
Relation	X

Historical Model

Attribute	Value
World History	Check Below
Personal History	N/A
Player History	N/A

Krystian “Krakk” Klikowicz

Cognitive Model

Attribute	Value
Self-Esteem	0.5
Locus of Control	0.8
Cognitive Flexibility	0.4

Social Personality Model

Attribute	Value
Assertion	0.3
Conformity / Independence	0.9
Empathy / Self Focus	0.8
Ideology	Libertarianism
Wildcard	Inferiority
Faction	The Runner Networks
Social Position	Boss
Relation	X

Historical Model

NPC Dialogue Processing Pipeline with Stage Products

This pipeline defines how NPCs interpret player dialogue and generate responses, ensuring behavior is consistent, personality-driven, and world-aware.

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0. Pre-Input: World Knowledge Initialization

Purpose: Ground NPC behavior in objective context before any player interaction.

Components:

- **World History:** Knowledge of major events, wars, factional conflicts, political tensions.
- **Factional Alignment:** Position relative to other groups, allies, and enemies.
- **Personal History & Achievements:** Formative experiences shaping perception.

- **Player History:** Track of player's actions, reputation, past interactions relevant to NPC.

Product / Output:

NPC World Model – a structured knowledge base representing the NPC's worldview, factional biases, and relevant personal history.

This is the reference framework for interpreting all dialogue.

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1. Input: Player Dialogue

Purpose: Capture and structure the player's communication for NPC reasoning.

Components:

- **Choice:** Player's selected dialogue option.
- **Language Art:** Tone of dialogue (Challenge, Diplomatic, Empathetic, Manipulative, Neutral).
- **Contextual Reference:** Mentions of past events, factions, or player reputation.

Product / Output:

Structured Dialogue Vector – a data object containing parsed dialogue traits, tone, and contextual references.

This is the "raw data" the NPC interprets in the next stage.

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2. Cognitive Filtering / Belief Formation (Id Logic)

Purpose: NPC interprets raw input and forms an internal understanding or belief about the player's message.

Process:

- Apply **Cognitive Model**:
  - **Self-Esteem:** Confidence in opinion formation.
  - **Locus of Control:** Whether statements are seen as actionable or external.
  - **Cognitive Flexibility:** Openness to revise perceptions.
- **Bias & Distortion:** Focus on information aligning/conflicting with prior beliefs.

- **Belief Scoring:** Track certainty of interpretation.

Product / Output:

Belief State Object – a structured representation of what the NPC believes about the player's intent and content of the dialogue, including confidence levels.

This forms the cognitive "truth" from which social behavior emerges.

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3. Social Personality Congruence / Desire Formation (Superego Logic)

Purpose: Translate cognitive interpretation into socially-informed **desired outcomes** reflecting personality, social norms, and narrative role.

Process:

- Apply **Social Personality Model**:
  - **Assertion, Empathy/Self-Focus, Conformity/Independence, Ideology, Wildcard**
- Formulate **desires or goals** for the conversation:
  - Persuade, defend, build rapport, protect secrets, act cautiously.
- Assign **weighted priorities** to goals.

Product / Output:

Goal Vector / Desired Outcome List – a ranked and weighted list of social objectives for the NPC, guiding tone, phrasing, and action in dialogue.

This ensures personality and factional consistency influence the response.

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4. Output / Ego Mediation

Purpose: Convert internal beliefs and social desires into a concrete, game-facing dialogue.

Process:

- Merge **Belief State** + **Desired Outcomes** + **World Model** to select:
  - **Tone / Language Art**

    - **Content / Dialogue Line** consistent with history, faction, and personality
    - **Subtext / Bias** (quirks, sarcasm, ideology)
- Optionally adjust **relation score** with player based on alignment between response and player action.

Product / Output:

Final Dialogue Response – a complete NPC line (or lines) with tone, style, and content ready for display in the game.

Can be dynamically generated using an LLM or prewritten options, while staying consistent with NPC personality and beliefs.

Layman's Pipeline With Products

Stage	Product / Output	Purpose
0. World Knowledge	NPC World Model	Baseline beliefs and biases for all interactions
1. Player Input	Structured Dialogue Vector	Raw player data for NPC to interpret
2. Cognitive Filtering	Belief State Object	NPC's internal understanding of player input
3. Social Personality	Goal Vector / Desired Outcome List	Ranked social objectives shaping the response
4. Output / Ego	Final Dialogue Response	Complete in-game dialogue line with tone, style, and subtext

Layman's Pipeline

0. Is Pre-input. It is how the NPC knows who's in charge, who's at war and who's their ally or enemy.
1. Player speaks: The player could be friendly , challenging, neutral or manipulative. This may reference past events, other factions or the player's reputation. **The NPC reads this as raw data.**

2. **NPC Thinks.** The NPC processes what was said:
    a. What fits their belief.
    b. How confident they are in their opinions.
    c. how open they are to changing their mind.

    The NPC decides do i agree / care ?
3. NPC decides what they want. They think about what they want to achieve and how to act based on their personality. This is the stage where their personality really shapes the response.
4. The NPC responds. They choose a tone, craft a message and adjust subtly for emotions, quirks and hidden agendas.

World Context

- For validity, id add a background model. This just contains personal history which is only for characters who need immersive dialogue. It'd be a more objective, with 0 subjective bias model which then feeds the character: 2502 • Azraelean-Seraphimean war commences in Angelia. • Political and social conflicts are unreconciled, triggering the war. • Cherubians and Metatronians are displaced, becoming traveling asylum seekers. • A Cherubian Archprophet, remembered as The Envisioned, prophesises a land in the Sky that will given onto the Cherubians and taken from them. She is taken as a cynic. 2510 • War begins to halt; both sides near dead. • Angelia is in near ruin. • Humans hunt Cherubians/Metatronians. • Cherubians begin plans to build an interconnected city in the sky. • Human scientists begin to take technology from abandoned warfields to analyse an integrate into their own countries. They cannot understand the technology despite deconstructing it. 2533 • Angelia wiped out; land divided among existing countries. • Azraeleans and Seraphimeans disappear. • Cherubians construct Land of Hearts in the sky; Metatronians build Land of Youth below in the ocean. • Passiotechnology is brought to the Land Of Hearts, using Passion-cores to bring sustainable energy to Romance City. 2540 • Cherubians and Metatronians swap small teams of architects, engineers, and scientists to

strengthen each other’s cities. • Humans learn of the segregated winged lands; demand inclusion to prevent potential war. • Tensions spark first ideological factions among humans. Some sympathetise with the Cherubians and agree to their bureaucracy calling themselves the State Of Hearts. Their political human rivals the Romancian Isola. 2544 • Moore (Metatronian) and Vick (Cherubian) engineer and supercharge the Heart of Passion (HoP), sacrificing part of their life force and linking bloodlines to it. • Early Cherubian council formalizes management protocols for HoP maintenance; minor factional disagreements arise among council members over human inclusion. The Romancian Isola begins collecting support from The Lands Below, secretly amassing political influence. 2549 • Moore unknowingly has a child with a human woman. • A famous artist-whistleblower known as Jean Pope Chautlier dies in the Lands Below trying to publicise against the Romancian Isola’s regime. His Friends escape to The Lands Above. 2580 • Land of Hearts and Romance City established. • After nine months, humans demand political power. • First local artist collectives form, promoting communal recovery after city establishment; early Runners emerge as misfit street youth testing urban traversal. 2585 • Humans usurp 30% of political power; some Cherubians collude. • Tensions rise between humans and Cherubians. • Vick leaves to Land of Youth (Dream City) for unspecified reasons. • Addition: Cherubian council fractures over human integration; some factions quietly begin safeguarding Passion Cores for potential political leverage. 2599 • Humans align with lands below to overthrow Cherubian rule. • Many Cherubians displaced or killed; few integrated into Romance City under human governance. • Early factional ideologies form: Artists (Commonman) begin emphasizing collective restoration. Many small factions are littered around Romance City but the Romancian Government controlled by the Romancian Isola scare many factions into hiding. 2600s • Human rule over Romance City established; mismanagement emerges. • HoP becomes unstable; Cherubian engineers rare and irreplaceable. • Social and economic instability fuels the “passion problem”; small districts begin displaying signs of apathy and early Dysphoria. • Disenfranchised youths start running around the rooftops of Romance City, leaving graffiti as “proof-of-life.” 2679 • Alexa Moore-Vale, great-granddaughter of Moore, has a child with strong Metatronian genes, emotionally receptive. • Morisson Moses observes political corruption within the military. He murders a High-ranking official for his crimes against humanity

then flees into hiding. • The first informal coalition of Artists and Runners begins communicating quietly with each other, aware of growing instability. 2690 • Artists form union, the Commonman, preaching that mutual harmony is necessary to prevent societal collapse. • The Commonman begins using graffiti, music, and performances as ideological influence; districts slowly reflect faction-specific cultural markers. • Morisson Moses launches a charismatic smear campaign against the Romancian Isola Controlled Government. Many defect to his side. Tensions rise in the Land of Hearts massively. Civil War is imminent. 2692 • Vick's descendant bears a son: Kupid. 2698 • Kupid nicknamed Kid Kupid; locally renowned for strong Cherubian genes. • Learns early in life he must travel to obscure his existence from hunters. • 19-year-old Metatronian in Romance City begins being overwhelmed by emotional receptivity. 2703 • Serotonin Junkie steals the HoP. • Romance City begins slow decay; Artists & insurgents attempt to intervene for the "good" of the land. • Initial faction propaganda campaigns begin, reinforcing fear, hope, and mistrust across districts. 2705 • Dysphoria begins plaguing the Land of Hearts; people grow apathetic, weak, and tired. • Society polarizes further. 2708 • Runners train street kids to become a rooftop army; plan to reclaim HoP. • Hidden urban traversal routes and temporary graffiti messaging used as communication and training networks. 2712 • Kid Kupid travels over Romance City; catches Dysphoria but only loses wings. • Must understand and resolve conflicts tied to Cherubian bloodline. • Early factional response to Kupid's arrival begins;Artists revere him, Runners indifferent, Insubordinates wary but dependent. 2502 – Outbreak of War • The Azraelean-Seraphimean War ignites in Angelia, fueled by unresolved political, social, and economic tensions. Both nations are militarized and entrenched; conflict becomes all-consuming. • Cherubians and Metatronians, historically intermediaries and peacekeepers due to their wings and arcane heritage, are targeted and forced to flee; they become traveling asylum seekers. • A Cherubian Archprophet, The Envisioned, foresees a "land in the sky" that Cherubians will inhabit but ultimately lose control of—a vision dismissed by many as cynical. 2510 – War's Collapse and Migration • Both sides are near annihilation; Angelia is largely ruined. • Humans, fearing the winged, hunt Cherubians and Metatronians actively. Many are killed or enslaved. • Cherubians secretly plan the Land of Hearts, an interconnected city in the sky meant to preserve their culture and passion-based technologies. • Human

scientists attempt to reverse-engineer abandoned Cherubian/Metatronian technology. Unable to understand the devices, they collect artifacts for prestige and speculation rather than practical use. 2533 – Founding of Segregated Cities • Angelia wiped out; lands below are divided among surviving human states. • Azraeleans and Seraphimeans vanish entirely. • Cherubians begin Land of Hearts, a sky city using Passion-Cores, early energy sources that harness emotional resonance to power machinery and basic city functions. • Metatronians construct Land of Youth in the ocean, specializing in experimental sensory and emotional manipulation. 2540 – Interfaction Collaboration and Political Tension • Cherubians and Metatronians exchange small teams of engineers, architects, and scientists to strengthen cities; a rare moment of collaboration. • Humans become aware of these segregated winged cities; they demand inclusion to prevent potential conflict. • Ideological factions form among humans: ◦ State of Hearts (SoH): sympathetic to Cherubians, promote bureaucracy and order. ◦ Romancian Isola (RI): rival human faction, secretly amassing political influence, aligned with interests from the Lands Below. 2544 – Creation of the Heart of Passion • Moore (Metatronian) and Vick (Cherubian) engineer the Heart of Passion (HoP), sacrificing parts of their life force and linking their bloodlines. • Cherubian council formalizes HoP maintenance protocols; factions clash over human inclusion. • RI covertly gathers influence via Lands Below, ensuring humans gain leverage over Cherubian governance. 2549 – Death of Jean Pope Chautlier • Moore unknowingly fathers a human child, a genetic precursor for Metatronian line sensitivity. • Jean Pope Chautlier, artist-whistleblower, dies attempting to expose Romancian Isola corruption. • Surviving followers escape to the Land of Hearts, laying the groundwork for the Commonman: ◦ Ideology emphasizes collective restoration, empathy, and protection of weak or disenfranchised citizens. ◦ The faction preserves cultural memory and passion, acting as moral and social stewards. 2580 – Establishment of Romance City • Land of Hearts and Romance City fully established; humans demand political representation nine months later. • Early artist collectives form, directly influenced by Chautlier's ideology. • Proto-Runners emerge: street youth experimenting with urban traversal, graffiti, and agile mobility. 2585 – Political Fractures and Corruption • Humans secure 30% political power, aided by colluding Cherubians. • Cherubian council fractures; some factions safeguard Passion-Cores as potential political leverage. • Vick

leaves for Land of Youth, reasons unknown. • Social unrest grows; human overgovernance emerges, foreshadowing McCarthy-esque crackdowns against dissenters. 2599 – Overthrow Attempts & Factional Seeding • Humans align with Lands Below to attempt Cherubian overthrow. • Many Cherubians are displaced or killed; survivors integrate under human governance. • Factional ideologies crystallize: ◦ Artists (Commonman): defend Chautlier's communal vision. ◦ Runners: continue informal rooftop network, fostering emergent subculture. ◦ Insurgents: early Morisson Moses loyalists, frustrated by military corruption, begin forming underground coalitions. 2600s – "Passion Problem" Emerges • Human mismanagement entrenches; HoP instability begins. • Apathetic districts and early Dysphoria emerge; disenfranchised youths express identity through graffiti and physical agility (proto-Runners). • Artists focus on medical and sensory passiotechnology, Runners focus on mobility and reconnaissance, Militants focus on defensive / offensive passiotech. 2679 – Moses Defection and Insurgency Begins • Alexa Moore-Vale gives birth to a child with strong Metatronian genes. • Morisson Moses, a high-ranking Romancian military officer, discovers corruption and abuses of power; kills a high-ranking official to protect civilians, flees into hiding. • Moses begins organizing the Insubordinates, a military utilitarian faction aiming for social order by sacrificing few for many. • Informal alliances form between Artists and Runners, sharing intelligence on HoP instability and rising Dysphoria. 2690 – The Commonman Solidifies • Artists formally unify under Commonman, following Chautlier's doctrine: communal stewardship, empathy, moral leadership, and holistic recovery. • Propaganda: murals, music, and performances propagate ideological messages. • Morisson Moses launches smear campaigns against RI, recruiting defectors. Tensions escalate; civil war looms. 2692 – Kupid's Birth • Vick's descendant fathers Kupid, a Cherubian human hybrid. 2698 – Kid Kupid Emergence • Kupid, known as Kid Kupid, becomes locally renowned for strong Cherubian blood. • Trained from youth to obscure his presence from hunters. • Metatronian in Romance City overwhelmed by extreme emotional sensitivity (prelude to Serotonin Junkie's rise). 2703 – Serotonin Junkie's Theft of HoP • Serotonin Junkie steals HoP; initiates societal decay. • Artists and insurgents attempt intervention. • Factional propaganda escalates: fear, hope, and ideological framing intensifies polarization. 2704 – The Fall of the Romancian Government • Serotonin Junkie, using full-capacity passiotechnology, systematically attacks

Romancian Isola politicians, targeting key decision-makers, bureaucrats, and ceremonial leaders. • Politicians flee the city; administrative collapse spreads. • Romancian Military is attacked by SJ's passiotech-enhanced assaults; human soldiers are defenseless against Metatronian energy manipulations. • Three-Day War: ◦ SJ's campaign destroys RI infrastructure. ◦ Morisson Moses 3rd parties military units; soldiers must defect to Insubordinates or die. ◦ Runners occupy abandoned districts, creating secure zones for civilians. ◦ Artists do the same, establishing cultural sanctuaries and medical/sensory relief hubs. 2705 – Dysphoria Spreads • Dysphoria becomes epidemic; citizens weaken emotionally and physically. • Social and political fragmentation deepens; factions further entrench. 2708 – Runners Mobilize • Runners train street youth into a rooftop army to reclaim HoP. • Hidden traversal routes, temporary graffiti messaging, and covert urban tactics develop into systematic underground communication networks. 2712 – Kid Kupid Arrives • Kid Kupid flies over Romance City; catches Dysphoria but loses only wings. • Must navigate political, emotional, and factional complexities tied to Cherubian bloodline. • Factional reactions: ◦ Artists: revere him as a moral and spiritual figure, protector of Chautlier's ideals. ◦ Runners: indifferent but pragmatic, monitoring potential disruption. ◦ Insubordinates: wary but reliant on Kupid for HoP recovery and city stability. We'd also in his social model align him to Faction: Insurgent Miltia*